

Workload, Fatigue, and Readiness in Elite Women's Rugby Sevens

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Abstract—This project studies whether recent workload measures are associated with wellness outcomes in the Canadian National Women's Rugby Sevens dataset. Using player-reported wellness, perceived exertion, and game context data, I examined relationships between workload and fatigue, monitoring score, and training readiness. The results showed that workload variables were meaningfully related to wellness, with training readiness giving the strongest model fit. In particular, the acute-chronic workload ratio and game-day status were consistently important. Overall, the findings suggest that workload metrics can help support athlete monitoring, but the relationship between load and fatigue is complex.

I. INTRODUCTION

Fatigue management is important in high-performance sports because excessive workload can reduce performance and increase injury risk. Rugby Sevens is especially demanding because athletes may play multiple games within a short time period. This report focuses on one question: whether recent workload measures are associated with athlete wellness and readiness.

GitHub repository for R code, figures, tables, and data: <https://github.com/ssyht/BigDataAnalysis-FinalProject>

II. DATA AND METHODS

The data come from the 2017–2018 Canadian National Women's Rugby Team season and use three files: `wellness.csv`, `rpe.csv`, and `games.csv`. The files were merged using `Date` and `PlayerID`, and players 18–21 were removed because they did not play in the provided season data.

The main outcomes are `Fatigue`, `MonitoringScore`, and `TrainingReadiness`. The main predictors are `AcuteLoad`, `ChronicLoad`, and `AcuteChronicRatio`. Because the RPE data contain multiple sessions per player per day, the workload data were aggregated to the player-day level and merged with the wellness data. A game-day indicator was also created from the games file. Linear regression models with player fixed effects were fit:

$$Y_{it} = \beta_0 + \beta_1 \text{AcuteLoad}_{it} + \beta_2 \text{ChronicLoad}_{it} + \beta_3 \text{ACR}_{it} + \beta_4 \text{GameDay}_{it} + \alpha_i + \varepsilon_{it} \quad (1)$$

where Y_{it} is a wellness outcome for player i on day t .

III. RESULTS

Fatigue was concentrated around moderate values, monitoring score was centered in the high teens, and training readiness

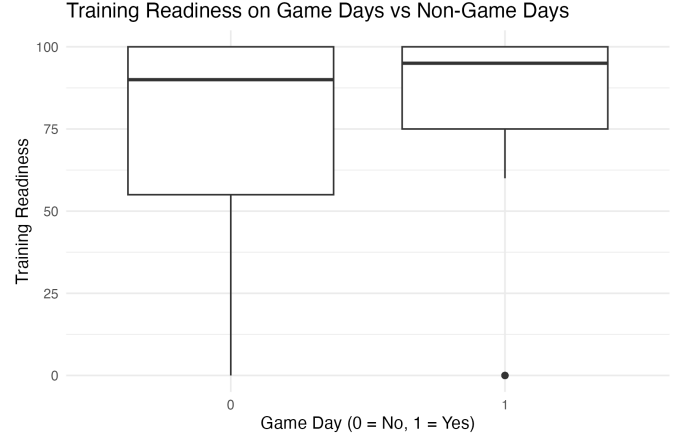


Fig. 1. Training readiness on game days versus non-game days. Readiness appears slightly higher on game days.

was generally high. Across the three models, workload variables were associated with wellness outcomes, although the direction and strength of the relationships varied by outcome. The fatigue model had an R^2 of about 0.35, the monitoring score model had an R^2 of about 0.37, and the training readiness model had the strongest fit with an R^2 of about 0.54.

The acute-chronic ratio was one of the most consistent predictors across all three models. Game-day status was also significant across the models. Figure 1 compares training readiness on game days and non-game days. The boxplot suggests that readiness was slightly higher on game days, which matches the positive game-day effect in the regression model. For fatigue and monitoring score, acute load showed a slight negative relationship in both the exploratory plots and regression results, suggesting that workload was related to athlete wellness, but not always in the simple direction of “more load means worse outcomes.”

IV. DISCUSSION AND CONCLUSION

The main finding of this project is that workload metrics were meaningfully associated with athlete wellness, but the relationship was complex. Training readiness was the strongest outcome in the models, and the acute-chronic ratio and game-day status appeared to provide useful information about how prepared athletes felt. At the same time, fatigue and monitoring score behaved somewhat differently, suggesting that these outcomes may capture different dimensions of athlete condition.